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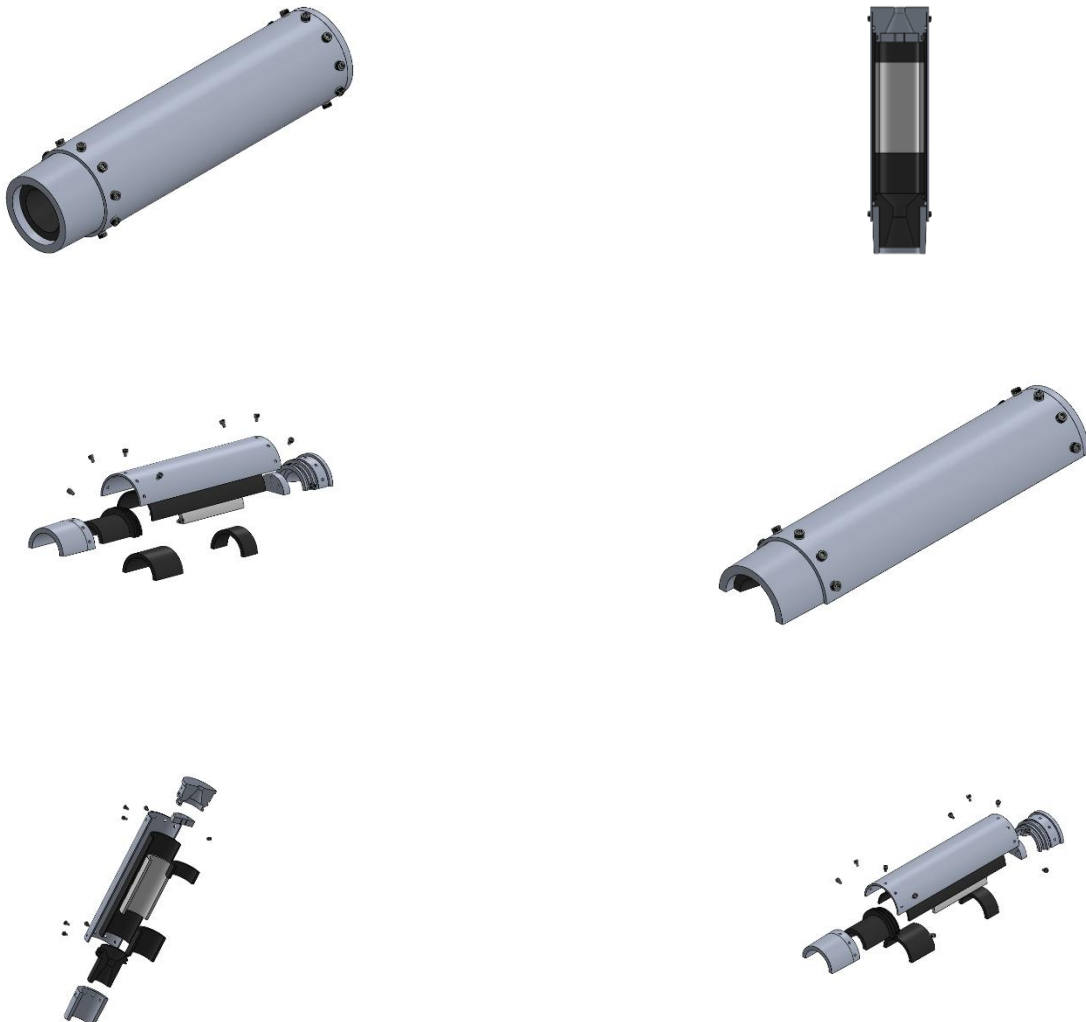
Relevant Technical files stored on Github: <https://github.com/nrobertson1453/portfolio>

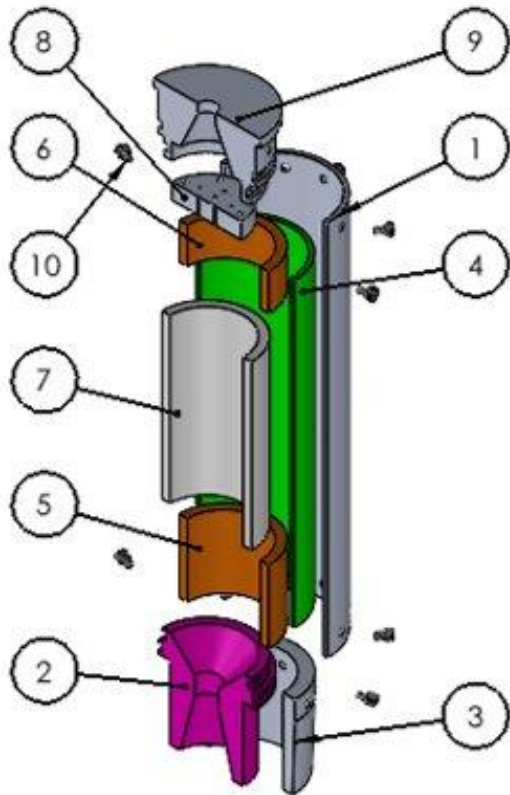
Combustion Chamber, Hybrid Rocket Motor N-class and K-class

Fall 2024-Fall 2026

- Contributed to the design and development of the combustion chamber of an N-class hybrid rocket motor (20,000 N-s of total impulse) Fall 2024 and Spring 2024
- Fall 2025 spearheaded the design of a K-class motor (2,500 N-s of total impulse). Using lessons learned from N-class motor, led extensive and iterative design decisions regarding safety, thermal and structural analysis, optimization, material selection, and fabrication, all within tight time and cost constraints, in close coordination with several other design teams. Produced a complete CAD assembly for the K-class motor.
- Utilized a team Matlab script to determine bolt placement, number, kind, and resultant stresses. Python was similarly used to determine O-ring size, groove depth and width, as well as compression. Code found in Github.

K-class Combustion Chamber CAD (SOLIDWORKS Assembly files in GitHub):





BOM Table		
ITEM NO.	PART NUMBER	QTY.
1	Casing	1
2	Nozzle	1
3	Nozzle Casing	1
4	Fuel Grain Liner	1
5	Post Combustion Chamber Liner	1
6	Pre Combustion Chamber Liner	1
7	Fuel Grain	1
8	Injector Plate	1
9	Injector	1
10	Bolts	20

PEP Racer: Potential Energy Powered Racer

- Designed, tested, and manufactured a “mouse-trap” style car utilizing propellers, springs, and a ramp with strict mass, part, and size constraints with team of 6. Singlehandedly redesigned car to function during iteration. Team scored one of the highest speeds in class of 168.



WheelWatch

Fall 2024

- Designed and developed over a 4-week period with a small team of 6 a system to alert bikers to free bicycle parking on Grounds from conception to manufactured prototype.
- Our design was a semi-circular indentation in the ground designed to fit the majority of human-powered bicycles. We attached a small force sensor connected to an Arduino which communicated with a website alerting bikers to the spot's availability.
- Project was designed to help the UVA community become more sustainable by encouraging commuters to choose to bike by ensuring they would have ample parking opportunities.
- Constructed from 3D printed parts designed in Autodesk Fusion, and wooden parts cut with a bandsaw.

Project Poster:



WheelWatch

Evan Daniel, Abel Saberhagen, Omer Mirza, Nolan Robertson, Jon Buszard, Austin Phan

SUMMARY

Our design addresses the issue of bikers not knowing whether a bike rack is available, which often discourages bike use and leads to bikes being locked in random areas. To overcome this problem, we created and manufactured a bike parking system that provides real-time updates on parking availability.

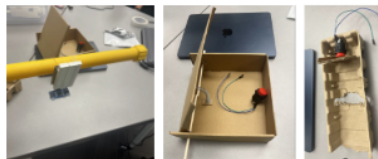
PROBLEM STATEMENT

The lack of secure and accessible bicycle parking on campus presents significant challenges for students commuting by bike. This shortage increases the risk of theft and discourages biking, pushing more students to drive, which raises carbon emissions and contributes to climate change (Macalester College, AASHE, 2024). If left unaddressed, students may face delays, reduced convenience, and less sustainable transportation options, leading to an overall decrease in quality of life and a negative perception of the university's support for sustainable practices. Ideally, a solution would provide lightweight, cost-effective, and easily producible parking infrastructure, ensuring that students can reliably and efficiently park their bikes near classes.

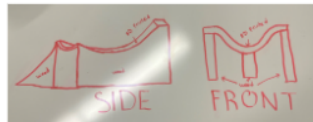
CITATIONS

He, F. (n.d.). Undergraduate Computer Students: Challenges and Struggles. UROC. Retrieved November 4, 2024, from <https://res.eric.ed.gov/fulltext/ED1341818.pdf>
AASHE. (n.d.). University of Virginia CP-16: Commute Model Spots: The Sustainability Tracking, Assessment & Rating System. Retrieved November 4, 2024, from <https://reports.aashe.org/track/sustainability/commute-model-spots/2024-03-04/20transportation/2016/>
Bike Theft Prevention - Public Safety. (n.d.). Macalester College. Retrieved November 4, 2024, from https://www.macalester.edu/public-safety/bike-theft-prevention/bike_theft_prevention/

DESIGN PROCESS



Creating our initial prototypes we found that the first prototype would be ineffective and ended up combining prototypes 2 and 3 for our final design. We decided that using force to detect the bike would be best due to the more compact and maintainable design. We kept the push down mechanic and the groove for the wheel. We assembled the arduino, we did several tests with the force sensor to determine if using force to detect the bike's presence was feasible. We then determined the width of the groove by collecting data on the average bike tire width around grounds. CAD models of the groove were then created using the collected information. We realized that there needed to be more stability for the groove, so we added wood boards that would be attached on the sides. The safety of the arduino was also taken into account, so a shell in CAD for the arduino was created to prevent environmental harm such as rain as well as the harm from the impact of physical force. The bike wheel however, needed to somehow get into the groove, so we added another part that would allow the bike to roll its wheels into place.



FUNCTION

The system features features a bike ramp that guides the bike into a bowl-like enclosure, securely encasing the tire. A sensor within the enclosure then detects the presence of the bike and sends a notification to a website, updating it to indicate that the spot is occupied.

FINDINGS AND FUTURE

Our project was ultimately successful in addressing the issue of inaccessible bike parking by developing a functional bike parking notification system. If we had more time or better materials, we would use a waterjet to produce more precise parts, incorporate additional 3D printing for better cohesion within the system, and fully integrate the system with the website wirelessly to enhance real-time functionality.

DESIGN CONSTRAINTS & CRITERIA

1. Our design must be user-friendly and be effective for those traveling to classes.
2. Bicycle users must be able to easily see the availability of a parking space before they get to class.
3. The design must be compact enough to fit within standard bike parking areas
4. The sensor and notification system must work consistently
5. The system should significantly improve bike parking organization and increase convenience.

Final Prototype Structure:



- Bridge**

 - Designed and manufactured a “bridge” able to hold 50 lbf for a class competition utilizing 2 in³ of 3D ABS plastic with a team of 6. Project involved the use of progressive design, CAD, material and structural analysis, and team coordination.

Fall 2025

Our final design and SolidWorks FEA analysis

